Question 2ai: Step-by-Step Explanation

ECO 310 Midterm Review

The Question

Show that someone prefers accepting (rather than rejecting) a risky bet if and only if their certainty equivalent for the gamble exceeds their initial wealth.

Key Definitions

Important Terms

- w_0: Initial wealth (what you have now before deciding on the bet)
- $u(w_0)$: Utility from keeping your initial wealth (rejecting the bet)
- **EU**: Expected utility from taking the risky bet
- **CE**: Certainty equivalent – the guaranteed amount that makes you indifferent to the gamble
- **Key relationship**: $u(CE) = EU$ (by definition of CE)

What We Need to Prove

We need to prove a **bi-directional** relationship (if and only if):

$$\boxed{\text{Person prefers accepting the bet} \Leftrightarrow CE > w_0}$$

This requires proving **two directions**:

1. **Forward direction (⇒)**: If they prefer accepting the bet, then $CE > w_0$
2. **Backward direction (⇐)**: If $CE > w_0$, then they prefer accepting the bet

Step-by-Step Proof

Direction 1: If prefer accepting bet $\Rightarrow CE > w_0$

Step 1: Start with the assumption

Assume the person prefers accepting the bet over rejecting it.

Step 2: What does this mean in terms of utility?

Preferring the bet means:

$$EU > u(w_0) \tag{1}$$

The expected utility from the risky bet exceeds the utility from keeping initial wealth.

Step 3: Use the definition of certainty equivalent

By definition, the certainty equivalent satisfies:

$$u(CE) = EU \tag{2}$$

Step 4: Substitute

From Steps 2 and 3:

$$EU > u(w_0) \tag{3}$$

$$u(CE) > u(w_0) \quad (\text{since } u(CE) = EU) \tag{4}$$

Step 5: Apply monotonicity of utility

Since utility is increasing in wealth (more money = higher utility), we know:

$$u(CE) > u(w_0) \implies CE > w_0 \tag{5}$$

Step 6: Conclusion

Therefore, if someone prefers accepting the bet, then $CE > w_0$.

Direction 2: If $CE > w_0 \Rightarrow$ prefer accepting bet

Step 1: Start with the assumption

Assume $CE > w_0$.

Step 2: Apply monotonicity of utility

Since utility is increasing in wealth:

$$CE > w_0 \implies u(CE) > u(w_0) \tag{6}$$

More wealth gives higher utility.

Step 3: Use the definition of certainty equivalent

By definition:

$$u(CE) = EU \tag{7}$$

Step 4: Substitute

From Steps 2 and 3:

$$u(CE) > u(w_0) \tag{8}$$

$$EU > u(w_0) \quad \text{(since } EU = u(CE)\text{)} \tag{9}$$

Step 5: Interpret the inequality

$EU > u(w_0)$ means the expected utility from the risky bet exceeds the utility from keeping initial wealth.

Step 6: Conclusion

Therefore, the person prefers accepting the bet over rejecting it.

Summary

The Complete Proof

We've proven both directions:

1. **Forward**: Prefer bet $\Rightarrow EU > u(w_0) \Rightarrow u(CE) > u(w_0) \Rightarrow CE > w_0$
2. **Backward**: $CE > w_0 \Rightarrow u(CE) > u(w_0) \Rightarrow EU > u(w_0) \Rightarrow$ Prefer bet

Key insight: The certainty equivalent acts as a threshold. If the "sure thing" value (CE) that makes you indifferent to the gamble exceeds what you currently have (w_0), then the gamble must be attractive enough to accept.

Intuition

Think of it this way:

- The CE is the guaranteed amount that makes you say "I'm indifferent between having this for sure vs. taking the gamble."
- If $CE > w_0$, it means you'd need to be given *more* than your current wealth to be indifferent to the gamble.
- But if you don't take the bet, you only get w_0 (your current wealth).
- Since the bet is "worth" CE to you (in terms of guaranteed value), and $CE > w_0$, the bet is better than keeping w_0.
- Therefore: $CE > w_0$ means the bet is attractive!